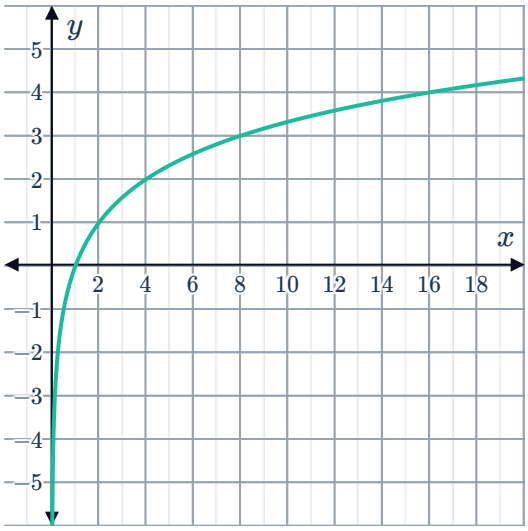
Logarithmic functions

1

a

x	$\frac{1}{2}$	1	2	4	16
y	-1	0	1	2	4

b



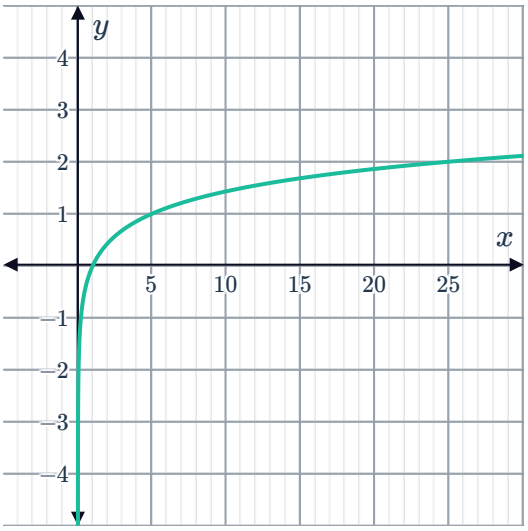
c $x = 0$

d

i No **ii** No **iii** Yes **iv** No

v No

2



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3

a

x	$\frac{1}{243}$	$\frac{1}{3}$	1	3	9	81
y	-5	-1	0	1	2	4

b Increasing

c $y = \log_3 x$ approaches $-\infty$ as x approaches zero.

d y is undefined.

Inverses of logarithmic functions

4

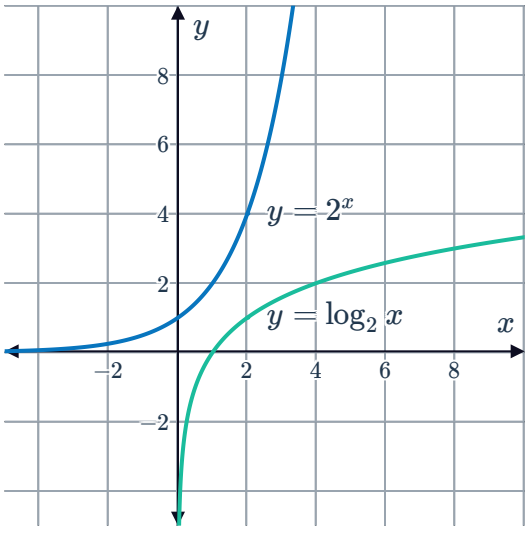
a

x	1	2	4	8
y	0	1	2	3

b

x	0	1	2	3
y	1	2	4	8

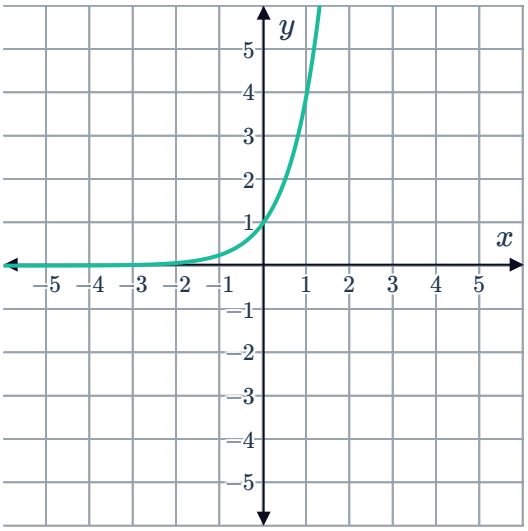
c



d $y = 2^x$

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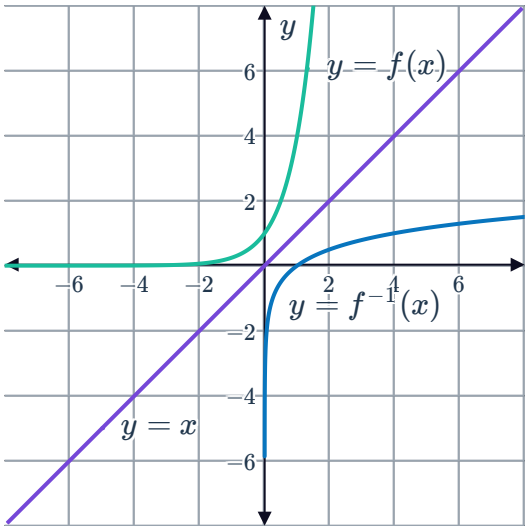
5



6

a

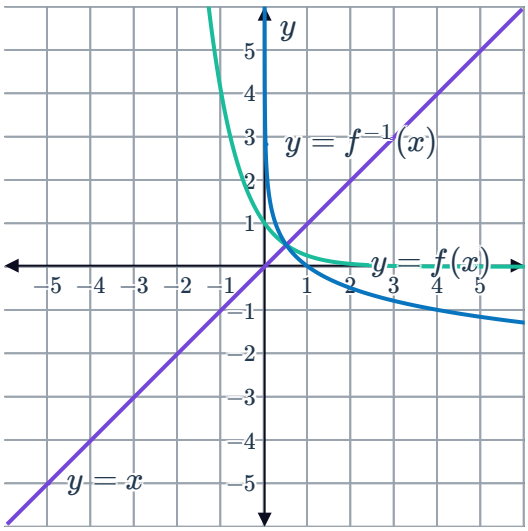
i



ii $y = \log_4 x$

b

i

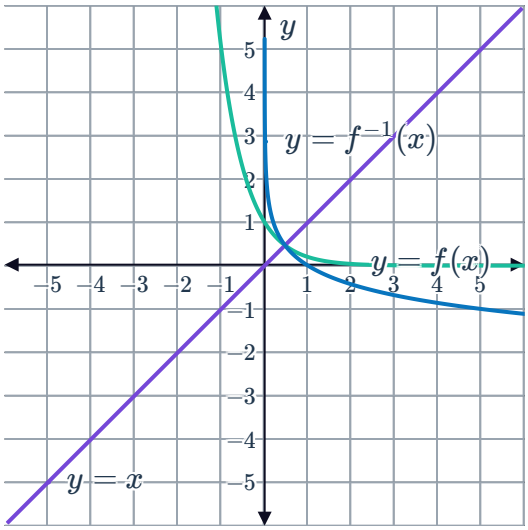


ii $y = -\log_4 x$

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c

i



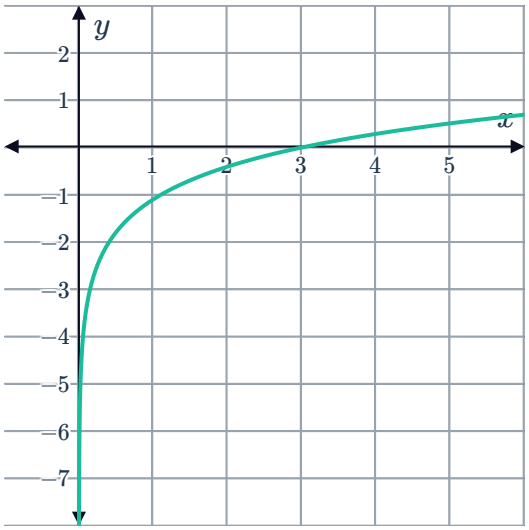
ii $y = \log_{\frac{1}{5}} x$

Natural logarithmic functions

7 The logarithmic function with base e.

8

a



b $y = \ln\left(\frac{x}{3}\right)$

9

a $y = \log_e\left(\frac{x+5}{2}\right)$

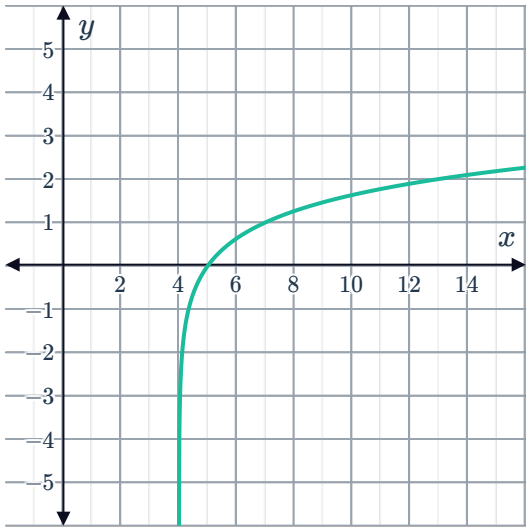
b $(-4.986, -4.986)\,,\ (1.118, 1.118)$

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Transformations of logarithmic functions

10 $y = \log_3 (x - 5)$

11

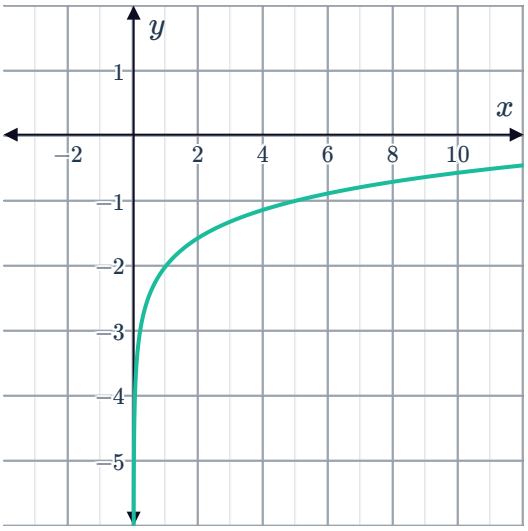


12

a

i $y = \log_5 x - 2$

ii

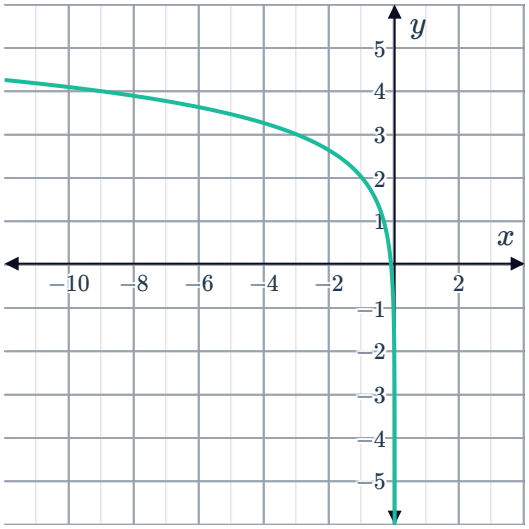


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b

i $y = \log_3(-x) + 2$

ii



13 $g(x) = 3\log_5 x$

14

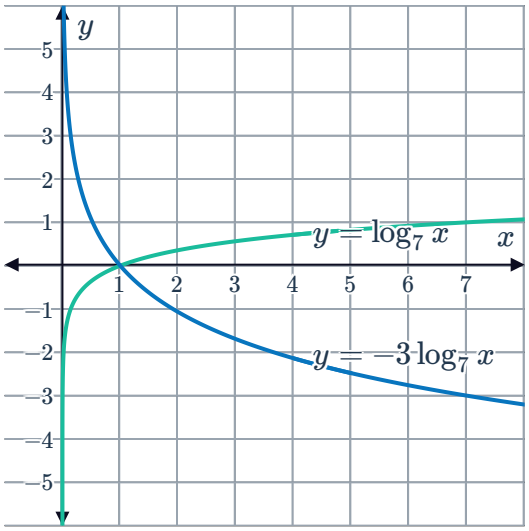
a Reflection over the y-axis.

b $g(x) = ln(-x)$

15

a Vertical stretch by a factor of 3 and reflection across the x-axis.

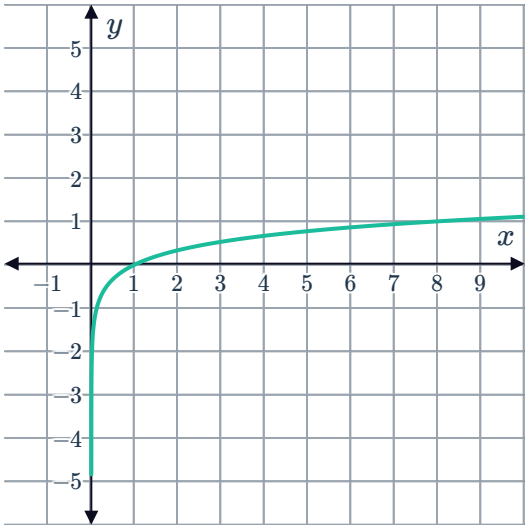
b



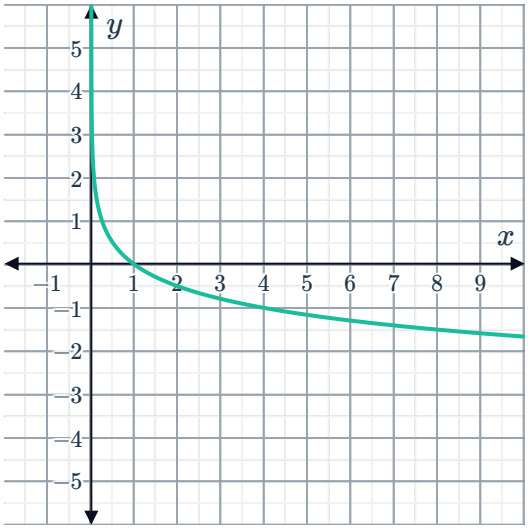
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16

a



b



17

a

x	-4	-1	$-\frac{1}{4}$
$\log_4(-x)$	1	0	-1

b

x	-2	1	$\frac{7}{4}$
$\log_4(2-x)$	1	0	-1

c $y = \log_4(2-x)$ is a translation of $y = \log_4(-x)$ by 2 units to the right.

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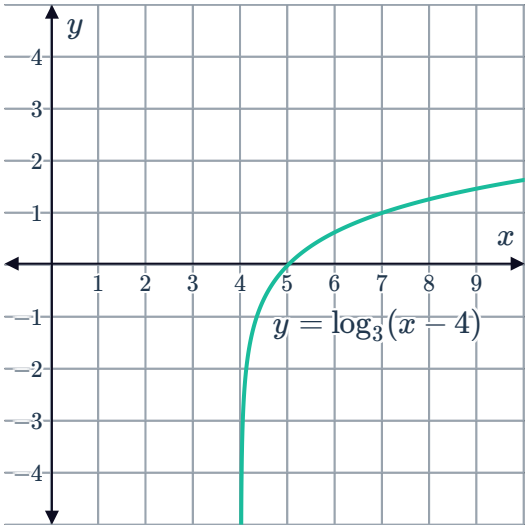
18

a $x = 4$

b $(5, 0)$

c 1

d



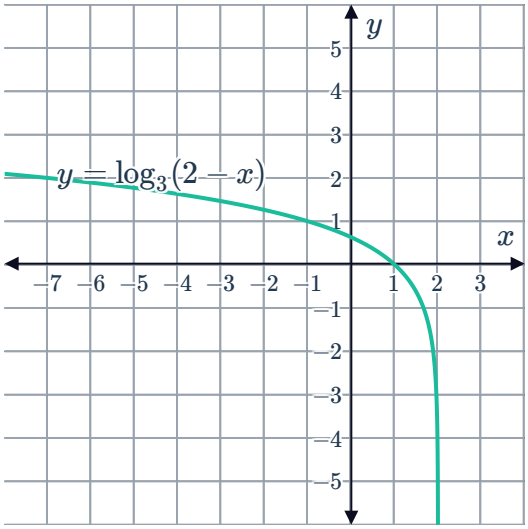
19

a $x = 2$

b $(1, 0)$

c 2

d



20

a $x = 0$

b $x = 0$

c $x = 5$

d $x = 0$

e $x = -3$

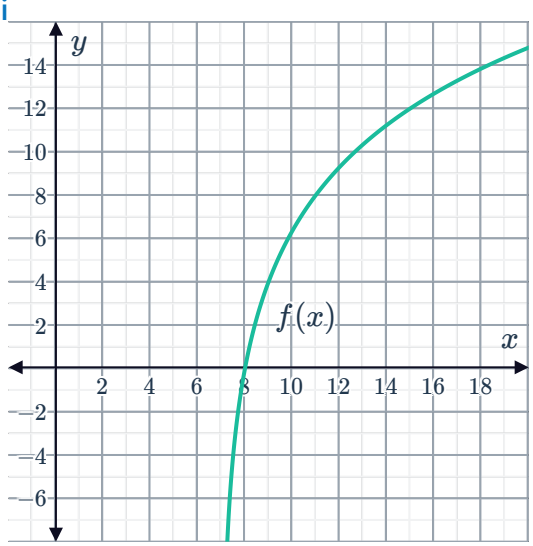
f $x = 0$

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21

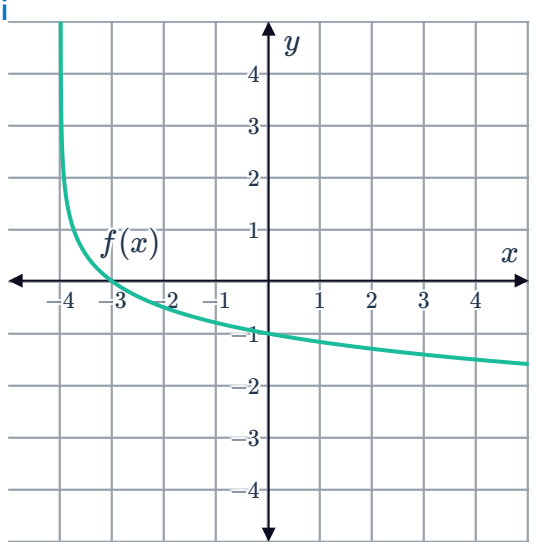
a

i $x = 8$ **ii** $x = 7$ **iii**



b

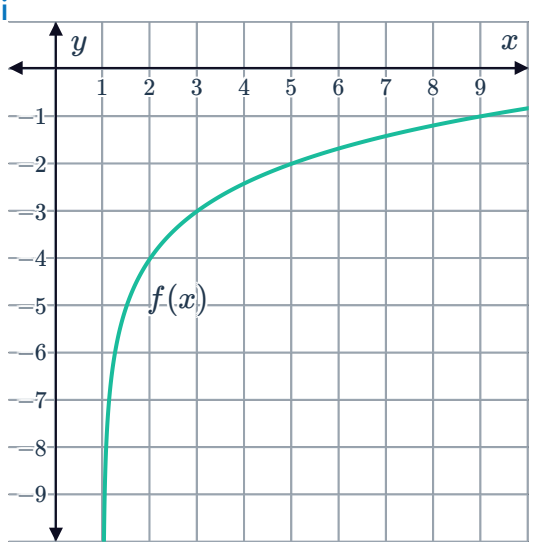
i $x = -3$ **ii** $x = -4$ **iii**



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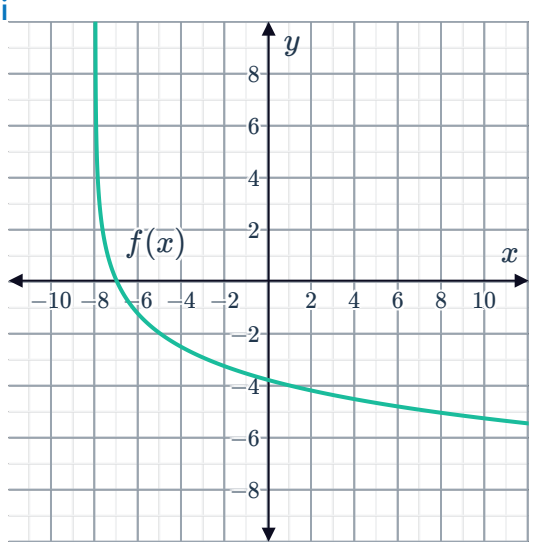
c

i $x = 17$ **ii** $x = 1$ **iii**



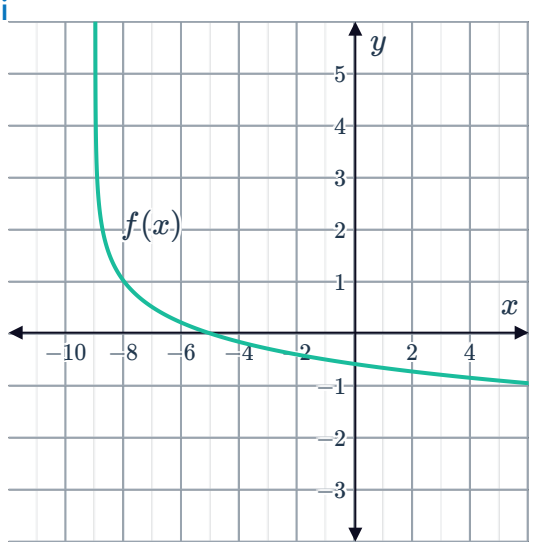
d

i $x = -7$ **ii** $x = -8$ **iii**



e

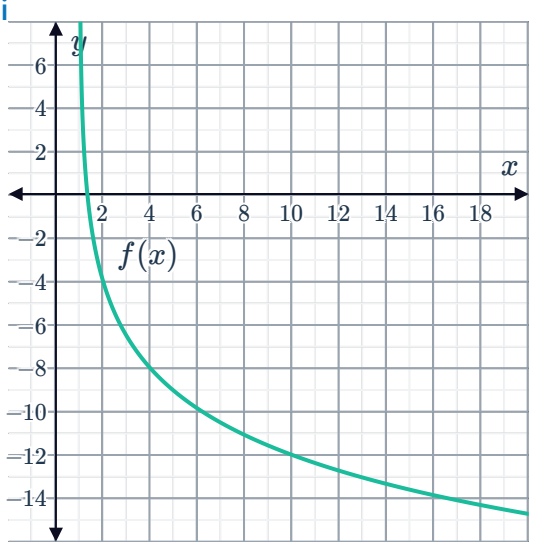
i $x = -5$ **ii** $x = -9$ **iii**



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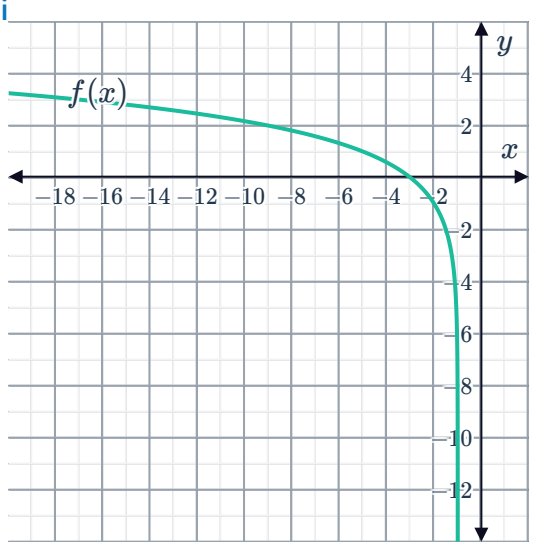
f

i $x = \frac{4}{3}$ ii $x = 1$ iii



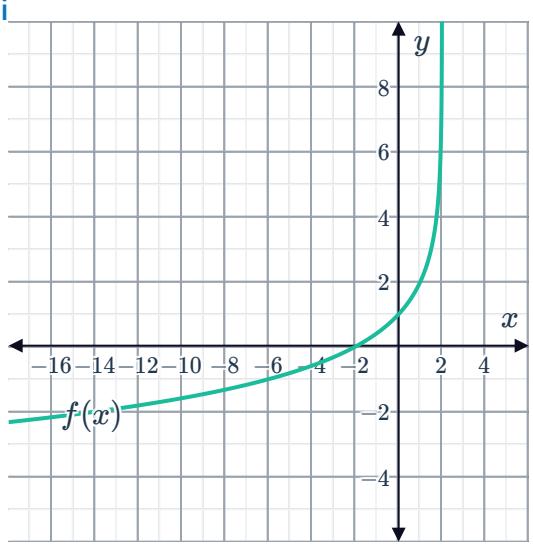
g

i $x = -3$ ii $x = -1$ iii



h

i $x = -2$ ii $x = 2$ iii



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Domain and range of logarithmic functions

22

a $f(x)$ has been translated 5 units to the right to give $g(x)$.

b The domain changes, but the range remains unchanged.

23

a

i $y = \frac{1}{5}e^x$

ii $(-\infty, \infty)$

iii $(0, \infty)$

b

i $y = \frac{1}{4}(e^x + 3)$

ii $(-\infty, \infty)$

iii $\left(\frac{3}{4}, \infty\right)$

c

i $y = \frac{1}{2}\left(e^{\frac{x}{4}} + 5\right)$

ii $(-\infty, \infty)$

iii $\left(\frac{5}{2}, \infty\right)$

d

i $y = \frac{1}{5}e^{\frac{x+3}{6}}$

ii $(-\infty, \infty)$

iii $(0, \infty)$

e

i $y = e^{\frac{x+3}{6}} + 4$

ii $(-\infty, \infty)$

iii $(4, \infty)$

f

i $y = \log_e x - 3$

ii $(0, \infty)$

iii $(-\infty, \infty)$

g

i $y = \log_e(x + 3)$

ii $(-3, \infty)$

iii $(-\infty, \infty)$

h

i $y = -\log_e(5 - x)$

ii $(-\infty, 5)$

iii $(-\infty, \infty)$

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i

i $y = \log_e\left(\frac{x}{2}\right) - 3$

ii $(0, \infty)$

iii $(-\infty, \infty)$

j

i $y = \frac{1}{5}\log_e\left(\frac{x-4}{3}\right)$

ii $(4, \infty)$

iii $(-\infty, \infty)$

24

a $x > 0$

b $x < 0$

c $x > 0$

d $x < 0$

e $x > -3$

f $x > -5$

g $x < 3, x > 7$

h $-4 < x < 0, x > 4$

i $x > -5$

j $x < 3$

k $-\infty < x < \infty$

l $-5 < x < 5$

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